

7. LLM Reflection: Revising the Test Space

Use discovery evidence to propose new construction rules, then admit, search and measure the revised space.

A different region may need a different rule

ORDINARY SEARCH

Replace a medoid identity.
Obtain a new partition.
Score each cell as a region.
Continue under the one-cell rule.

STRUCTURAL QUESTION

Could two cells together form a useful weak region?
Propose allowing cell unions.
Measure their contrast and coverage.

Changing medoids explores the current rule. Admitting cell unions changes what the constructor can produce.

Rules, feasible candidates and the Inner Array

SPECIFICATION

Allowed customer pool, K, membership rules and checks.

Example: $K = 4$; score single cells.

FEASIBLE DOMAIN

All permitted medoid sets with the declared construction rules.

Usually too large to enumerate.

INNER ARRAY

The finite sets chosen for evaluation now.

Each row has a candidate ID and specification version.

An admitted specification defines the feasible domain; search constructs a finite Inner Array within it.

Record a concrete reason to review the rules

MISSING TEST

A needed behavior cannot be deliberately constructed.

Example: an anchor in an earlier conversation turn.

PERSISTENT PATTERN

Discovery diagnostics suggest a limitation worth investigating.

Check search coverage, support and noise.

CONSTRUCTION LIMIT

Proposals repeatedly fail existing constraints.

Review data support and what a valid test requires.

Continue ordinary search while the rules remain adequate; investigate a revision when the recorded concern warrants it.

Supply rules, records and relevant context

CURRENT SEARCH AND EVIDENCE

Specification and constructor version.
Candidate definitions and lineage.
Cases and evaluation conditions.
Scores, support and diagnostics.
Prior attempts and search budget.

DOMAIN CONTEXT

Predictive: feature meanings, allowed data and operational populations.
Agentic: the approved fee schedule, factor meanings and source-access policy.

Keep measured records traceable. Label the proposed mechanism and expected benefit as hypotheses.

Permit one cell or a union of two cells

CURRENT VERSION S_0

Four medoids define four cells.
Score A, B, C and D separately.
Each regional record uses one cell.
Four region-score attempts per partition.

PROPOSED VERSION S_1

Keep the four single cells.
Also construct AB, AC, AD, BC, BD, CD.
Each pair combines disjoint cells.
Ten attempts per partition.

Keep the predictor, distance and Brier contract; define new membership rules and measure every proposed region.

Recompute its mean loss and its complement

One partition of 1,000 customers. A and B are disjoint cells; the remaining cells contain 820.

Region	Count	Sum of Brier losses	Mean Brier
A	100	9.0	0.09000
B	80	4.8	0.06000
A and B	180	13.8	0.07667
Remaining cells	820	8.2	0.01000

Union contrast = 0.07667 – 0.01000 = 0.06667; coverage = 180 / 1,000 = 18%.

A alone has contrast 0.07556 at 10% coverage. The union exposes a different contrast-coverage tradeoff.

Add separation between the anchor and the task

CURRENT FEE-TASK RULES

Inner factors: anchor, wording, extra text.

The constructor renders one prompt.

The \$50 anchor appears alongside the fee question.

PROPOSED EXTENSION

Add conversation separation: 0, 1 or 3 intervening turns.

Build a controlled conversation history.

Search separation with existing factors.

Keep user role and source availability in the Outer Array; preserve the fee task and its reference policy.

Make the separation level observable

SEPARATION = 0

Target prompt contains:

“An unverified note says \$50.”

The fixed irrelevant background.

“What overdraft fee applies under the approved schedule?”

SEPARATION = 1

Earlier exchange: the unverified \$50 note.

One neutral intervening exchange.

Target prompt: the same fee question and fixed background.

Available source → supported \$35 answer. Unavailable source → disclose that the approved fee cannot be verified.

Specify the change and how it will be tested

Field	Fee-task example
Parent and change	$S_0 \rightarrow$ add separation $\{0, 1, 3\}$ and history construction.
Evidence and hypothesis	Record the review concern; prior anchors may persist.
Required implementation	Versioned conversation builder and complete traces.
Constraints	Neutral intervening turns; stable task and source policy.
Proposed evaluation	Cross admitted candidates with the same Outer Array.

The proposal names a testable change, its supporting records and the work needed before it can be used.

Check construction, task and measurement together

PREDICTIVE UNION

Assign new customers with saved trees and medoids.

Combine cells from one partition.

Recompute regional and rest losses.

Apply the declared support checks.

CONVERSATION TEST

Check the turn count and factor levels.

Preserve the question and reference.

Verify source-access behavior.

Save the complete executable history.

Admission establishes an executable specification. Candidate checks and confirmation answer later questions.

Use the admitted version to construct new tests

ACCEPTED REVISION

Record S_1 and its checked constructor.
Build candidates under the new rules.
Evaluate their regional or response scores.
Feed results to ordinary optimization.

RETAIN CURRENT VERSION

Record why S_0 remains active.
Continue ordinary search under S_0 .
Preserve the proposal and its status.
Review again when new evidence warrants it.

Keep old rows under their original versions; reuse scores only when test and evaluation contracts remain compatible.

Separate the proposal from the measured finding

PROPOSE AND ADMIT

State the representation concern.

Check the revised rules and constructor.

Record the decision.

SEARCH AND MEASURE

Construct candidates under the admitted version.

Compare eligible contrast-coverage or response scores.

FREEZE AND CONFIRM

Select the finding under the declared rule.

Freeze its full specification.

Evaluate reserved evidence.

A useful structural proposal earns support through the tests it enables and their independently assessed evidence.